

# Sahil Malakar

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## SUMMARY

Final-year Electronics and Telecommunication Engineering student at Assam Engineering College with 2+ years of hands-on backend engineering experience across production systems. Skilled in building **distributed systems**, **microservices architectures**, and **concurrency-safe APIs** using Node.js, TypeScript, Prisma, and PostgreSQL — with real-world internship and freelance delivery experience. Actively seeking a full-time **Backend Engineer** role.

## TECHNICAL SKILLS

|                            |                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------|
| <b>Languages:</b>          | JavaScript, TypeScript, Python, Go                                                                |
| <b>Backend:</b>            | Node.js, Express.js, REST APIs, WebSockets, OpenAPI Spec, Rate Limiting, gRPC                     |
| <b>Databases:</b>          | PostgreSQL, MongoDB, Prisma ORM, Mongoose, Redis (caching, distributed locking)                   |
| <b>Frontend:</b>           | React.js, Next.js, Tailwind CSS                                                                   |
| <b>DevOps &amp; Cloud:</b> | Docker, AWS, CI/CD Pipelines, Cloudflare, Observability & Structured Logging                      |
| <b>System Design:</b>      | Microservices, Message Queues (BullMQ), Idempotency, Race Condition Handling, Scalable API Design |

## WORK EXPERIENCE

### Full Stack Developer (Freelance)

Jan 2024 – Apr 2025

Self-Employed · Node.js · TypeScript · PostgreSQL · Redis · Next.js · Docker · Socket.io

- Delivered multiple production full-stack applications for clients — including admin dashboards, e-commerce platforms, and **real-time systems using WebSockets** — independently end-to-end.
- Implemented **JWT-based authentication and RBAC** across client projects, and optimized database queries and indexing strategies, improving application performance by **~25%**.
- Deployed all applications using **Docker with CI/CD pipelines**, managing environments and ensuring zero-downtime releases for live client products.

## PROJECTS

### AI-Powered Placement Management System

Oct 2025 – Apr 2026

Node.js · TypeScript · PostgreSQL · Prisma · BullMQ · Redis · Docker |  
github.com/SahilMalakar/Ai\_powered\_placement\_management

- Architected a **modular MVC backend** (controller-service-repository) in Node.js + TypeScript, serving **500+ student profiles** with job postings and application records via REST APIs.
- Implemented **RBAC with JWT authentication**, an eligibility-check engine (CGPA, branch, backlogs), and **ATS-like resume scoring** — eliminating 100% of invalid job applications before any DB write.
- Built **async PDF generation** via BullMQ + Redis **message queue**, decoupling heavy processing from the request cycle and storing immutable profile snapshots to ensure **zero data drift**.
- Containerized with **Docker** for consistent CI/CD environments; added **centralized structured logging** and **rate limiting** on all public endpoints.

### SkyRoute — Distributed Hotel Booking Backend

Dec 2025 – Apr 2026

Node.js · TypeScript · Express · PostgreSQL · Prisma · Redis · BullMQ | github.com/SahilMalakar/SkyRoute

- Architected a **microservices-based** hotel booking platform in Node.js + TypeScript handling booking, inventory, user management, and notification services as **independent deployable units**.
- Solved **race conditions** in concurrent booking requests using a **Redis distributed locking** mechanism — ensuring exactly-one write per room slot and achieving **exactly-once booking semantics**.
- Built an **idempotency layer** to prevent duplicate charges on client retries, and implemented **BullMQ async job processing** for email/SMS notifications — fully decoupled from the booking path.
- Designed a **normalized PostgreSQL schema** optimized for read-heavy query patterns, with **centralized observability logging** across all services cutting debug time by **~40%**.

## EDUCATION

### Assam Engineering College, Guwahati

Oct 2022 – Aug 2026

B.Tech in Electronics and Telecommunication Engineering